



[SINGLE CORRECT CHOICE TYPE]

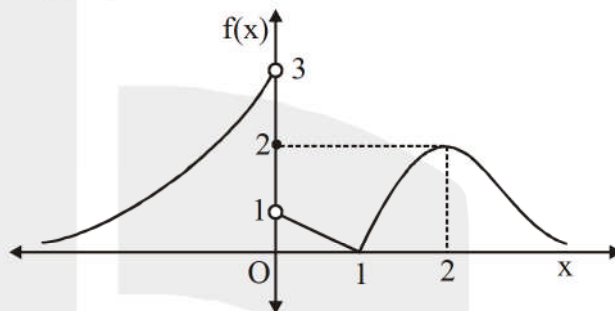
Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 If graph of a function $f(x)$ is shown in the adjacent figure, then

$$\lim_{x \rightarrow 0^-} \left[\frac{4f(x) - 6[f(x)]}{\tan(2f(x) - 6)} \right] \text{ is equal to}$$

Note : $[y]$ denotes the greatest integer function less than or equal to y .

- (A) 0
(B) 1
(C) 2
(D) 3



Q.2 A triangle is formed by points $P(2, 3)$, $Q(1, 1)$ and $R(t, t^2 + 4)$, then minimum area of ΔPQR is
(A) 2 (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) 4

Q.3 Let f be a differentiable function such that $f'(100) = 10$ then $\lim_{h \rightarrow 0} \left(\frac{f(100 + 3h^3) - f(100 - 7h^3)}{h^3} \right)$ is equal to
(A) 1 (B) 10 (C) 100 (D) 1000

Q.4 Let a, b, c be sides of a triangle ABC such that $\lim_{r \rightarrow \infty} \frac{\cos \frac{a}{r} - \cos \frac{b}{r} \cos \frac{c}{r}}{\sin \frac{b}{r} \sin \frac{c}{r}} = \frac{1}{2}$, then the value of

$$\frac{b^2 + c^2 - a^2}{bc} \text{ is equal to}$$

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 1

Q.5 Let $f : (-5, 5) \rightarrow \mathbb{R}$ be a differentiable function with $f(4) = 1$, $f'(4) = 1$, $f(0) = -1$ and $f'(0) = 1$.

If $g(x) = \left(f(2f^2(x) + 2) \right)^2$, then $g'(0)$ equals

- (A) 4 (B) -4 (C) 8 (D) -8



- Q.6 Let f be a continuous and differentiable function on $\mathbb{R}$ satisfying $f(-x) = f(x)$ and $f(2+x) = f(2-x) \forall x \in \mathbb{R}$ and $f'(1) = -5$. Then the value of $\sum_{r=0}^{100} (-1)^r f'(r)$ is equal to
 (A) 0 (B) -125 (C) 125 (D) 500
- Q.7 If $f(x) = \operatorname{sgn}\left(x^2 - kx + 6\right)\left(\sin x - \frac{1}{2}\right)$ (where $k > 0$) has exactly 4 points of discontinuity in $(0, 6)$ then maximum integral value of k is
 (A) 4 (B) 5 (C) 6 (D) 7
- Q.8 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $2f\left(\frac{x+y}{2}\right) - f(y) = f(x) \forall x, y \in \mathbb{R}$. If $f(0) = 5$ and $f'(5) = -1$, then
 (A) $\lim_{x \rightarrow 4} (f(x))^{\frac{1}{x-4}} = e$.
 (B) $f(|x|)$ is non-derivable at exactly 2 points.
 (C) area bounded by $f(x)$, x -axis and y -axis is 25 sq. units.
 (D) $|f(|x|)|$ is non-derivable at exactly 3 points.
- Q.9 Let $f(x)$ be a continuous function on $\mathbb{R}$ whose range is $[6, 7.5]$. If $h(x) = \left[\frac{3 \sin \alpha + 4 \cos \alpha + f(x)}{\lambda} \right]$, $\alpha \in \mathbb{R}$, $\lambda \in (0, 20]$, be continuous function then the number of integral values of λ is equal to
 [Note: $[k]$ denotes the greatest integer function less than or equal to k],
 (A) 0 (B) 8 (C) 9 (D) 13
- Q.10 If $a, b \in \{0, 1, 2, \dots, 10\}$ then number of ordered pairs (a, b) for which $f(x) = \left[\frac{a}{3}x + b \right]$ is continuous in $[0, 2]$ where $[k]$ denotes greatest integer less than or equal to k , is(are)
 (A) 10 (B) 11 (C) 20 (D) 22

[PARAGRAPH TYPE]

Q.11 & Q.12 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 & 12

$$\text{Consider, } f(x, y) = \lim_{n \rightarrow \infty} \frac{2^{nx+2} \sin^2(x-1+y) + \tan^2(x+2y) + \frac{2}{\pi}(x^2+y^2) \tan^{-1}(nx)}{2^{nx}(x-1+y)^2 + \lambda x^4 \operatorname{cosec}^2(1) + y^4}.$$

- Q.11 If $f(1, y)$ is continuous for all $y \in \mathbb{R}$, then the value of $f(1, 0)$ is

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) 2 (D) 4



Q.12 If $\lim_{x \rightarrow 0} f(x, 0)$ exists, then λ is equal to

- (A) $\frac{2}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{9}$ (D) $\frac{4}{3}$

[MULTIPLE CORRECT CHOICE TYPE]

Q.13 & Q.14 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.13 Let $f(x) = 2\sin^2\beta + 4\cos(x + \beta) \sin x \sin \beta + \cos 2(x + \beta)$ and $g(x) = \left(\lim_{\alpha \rightarrow 0} (1 + \alpha)^{\frac{1}{\alpha}} \right)^{\tan(\tan^{-1} x)}$.

Then

(A) number of integers in the range of $f(x)$ is 3.

(B) $\lim_{x \rightarrow 0} \left(f(x) \right)^{\frac{1}{x^2}}$ is equal to e^{-2} .

(C) $\lim_{x \rightarrow 0} \frac{f(x) - g(x)}{x}$ is equal to -1

(D) number of solution of the equation $g(x) = 1 - x$ is 2.

Q.14 If $y = \sqrt[3]{6xy^2 - 11x^2y + 6x^3}$, then possible value(s) of $\frac{dy}{dx}$ at $x = 2$ is(are)

- (A) 0 (B) 1 (C) 2 (D) 3

[MATCHING LIST TYPE]

Q.15 & Q.16 are **Matching List type** questions. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Q.15 Let $\lim_{x \rightarrow 0} \frac{a \sin x + bxe^x + 3x^2}{\sin x - 2x + \tan x}$ exists and has value equal to L.

List-I

P. The value of a is

Q. The value of b is

R. The value of L is

S. The value of $\frac{b-L}{a}$ is

List-II

1. -12

2. -3

3. 3

4. 6

Code :

(A) P-3, Q-2, R-3, S-1

(C) P-3, Q-2, R-1, S-3

(B) P-3, Q-2, R-3, S-3

(D) P-3, Q-3, R-1, S-3



Q.16 Consider $f(x) = \frac{-2x^4 + x^2 - 2}{x^4 + x^2 + 1}$ and $g(x) = x^2 + 2x + 3$.

List-I

- P.** If greatest value of $f(x)$ is M , then $|4M|$ equals
Q. The number of point(s) where $[f(x)]$ is discontinuous is (are)
R. The number of point(s) where $[g(f(x))]$ is discontinuous is (are)
S. The product of all real values of 'p' for which $\text{sgn}(g(x) + 2 - p^2)$ is discontinuous at exactly one point, is

List-II

- 1.** -4
2. 1
3. 2
4. 4

[Note : $[k]$ denotes greatest integer less than or equal to k and $\text{sgn}(k)$ denotes signum function of k .]

Code :

(A) P-4, Q-3, R-1, S-2

(B) P-2, Q-3, R-1, S-4

(C) P-3, Q-4, R-1, S-2

(D) P-4, Q-3, R-2, S-1

[INTEGER TYPE]

Q.17 to Q.20 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.17 If n_1 denotes number of points of non-differentiability of

$f(x) = \cos(2x - 1) \cdot \frac{\pi}{2} \cdot |x^2 - 4x - 5| + |x^5 + x^3| + x \cdot e^{|x|}$ and n_2 denotes number of points of discontinuity of $g(x) = \text{sgn}(\sin^2 x - \sin x - 1)$, $x \in (0, 4\pi)$. Find the value of $(n_1 + n_2)$.

Q.18 If $\tan^2 \alpha + \cot^2 \alpha = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 8x + 3} - \sqrt{x^2 + 4x + 2} \right)$ where $\alpha \in \mathbb{R}$, then find the value of $\sum_{n=1}^{\infty} (\sin^{2n} \alpha + \cos^{2n} \alpha)$.

Q.19 If f is a positive real valued differentiable function satisfying the functional rule

$f^2(x + y) = f^2(x) + f^2(y) - 2\sqrt{x} - 2\sqrt{y} + 2\sqrt{x + y} - 1$ for all $x, y \geq 0$ such that $\lim_{h \rightarrow 1} \frac{f(h) - 2}{h - 1} = \frac{1}{2}$, then find the number of solution(s) of the equation $\frac{f(|x|)}{\lambda|x| + 1} = 1$ ($\lambda > 0$).

Q.20 Let $y = f(x)$ ($f: \mathbb{R} \rightarrow \mathbb{R}$) be an explicit function defined by the implicit equation

$x^3 + y^3 + 3(x^2 + y^2) + 3(x + y) = 14$ and g be the inverse of f . If $\frac{d}{dx} (f(x + g(x)) \cdot g(x + f(x)))$ at $x = -1 + \sqrt[3]{15}$ is equal $\lambda(15)^{2/3}$, where $\lambda \in \mathbb{I}$, then find the value of $|\lambda|$.

ANSWER KEY

Q.1	B	Q.2	A	Q.3	C	Q.4	D	Q.5	D
Q.6	A	Q.7	C	Q.8	D	Q.9	B	Q.10	D
Q.11	D	Q.12	B	Q.13	ABC	Q.14	BCD	Q.15	C
Q.16	D	Q.17	4	Q.18	2	Q.19	3	Q.20	0

